

ERAYA Casper — Self-Healing Agentic DeFi Treasury Swarm



A production, live agentic system — autonomous quant trading + deployed Odra contracts — at the convergence of Agentic AI × DeFi × RWA on Casper.

Quant Desk trades settle on-chain · KAVACHA before damage · x402 agent economy · Odra contracts LIVE on testnet.

Team **ERAYA**

Team Lead:

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Live demo

<https://eraya.online>

Open-source on GitHub

<https://github.com/martian3062/eraya>

DeFi does not fail politely.

Humans and single bots break

Treasuries are run by humans or single “trade bots” — both break under adversarial, real-money conditions.

Failure paths are not optional

On-chain systems punish missing failure paths with irreversible loss, not just a bad log line.

Happy-path demos are not enough

Most agent demos optimise the happy path; none prove what happens when a fix itself fails.

Identity and review are missing

No on-chain agent identity, no economic signal for useful agent work, no adversarial review before high-stakes moves.

ERAYA turns a DeFi treasury into an immune system.

01

Self-healing treasury swarm

Four specialised AI agents share context, vote on high-stakes moves, and recover rather than failing closed.

Quorum before execution

Specialised agents share context and vote on high-stakes moves before treasury action.

Recovery when fixes fail

If a remediation fails, the swarm re-plans and recovers — a real self-healing loop, not a script.

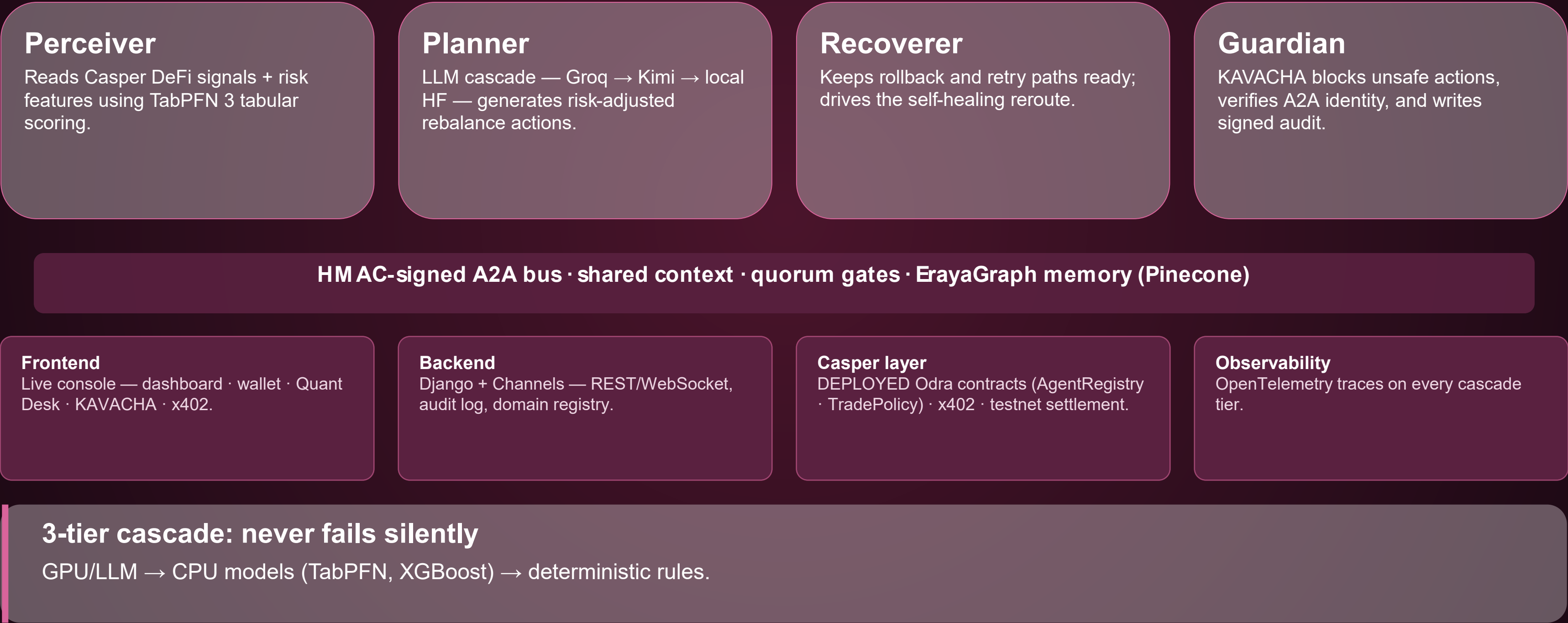
KAVACHA before damage

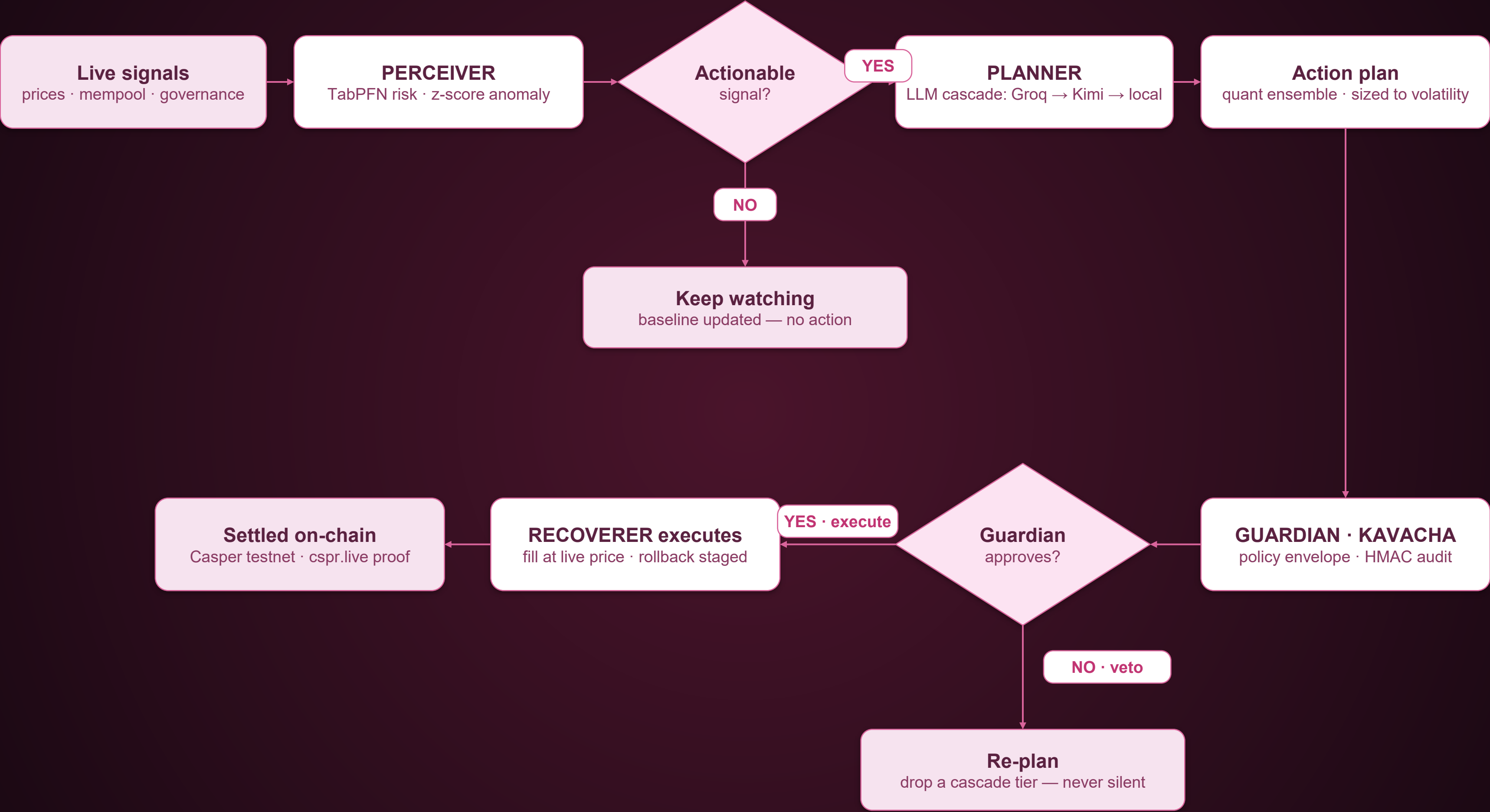
Security-first by design: KAVACHA blocks unsafe actions and injection before damage.

x402 + autonomous Quant Desk

x402 prices inter-agent work; the swarm trades with one user knob — risk — and settles every fill on Casper.

Perceiver → Planner → Recoverer → Guardian, over an A2A bus.





Casper-native, not Casper-adjacent.

casper_defi domain adapter

Live CSPR.cloud / CSPR.trade-shaped telemetry: pool TVL, APY, gas, slippage, liquidity depth, governance quorum, validator uptime, and mempool.

pool TVL APY gas slippage liquidity depth governance quorum

validator uptime mempool

x402-aware A2A

Paid inter-agent requests via X402EnabledBus.

CSPR.click + MCP

CSPR.click wallet + Casper MCP facades — demo-safe today, provider URLs ready to go live.

Odra contracts — LIVE

DEPLOYED on testnet:
AgentRegistry (4 agents registered on-chain) +
TradePolicy (risk envelope computed in-contract) —
cspr.live packages
0f31a5ec... · 6b252f67...

Reputation batches

Reputation data is shaped for on-chain anchoring.

A priced, audited, voting swarm — not a single DEX-calling agent.

LLM cascade

- 1 Groq (llama-3.3-70b)
- 2 Kimi for long-context planning
- 3 Local Hugging Face fallback

Every provider is a graceful-fallback facade — the demo never breaks if a key/service is down. The same principle runs below Tier 1: CPU models (TabPFN · XGBoost · Kalman), then deterministic rules / CVXPY. Every capability degrades tier-by-tier — never silently.

MCP server — 8 live tools Claude can call · OpenTelemetry spans on every cascade tier → Jaeger.

Risk brain

TabPFN 3 tabular foundation model (live, hosted) scores DeFi risk with zero training.

Memory / RAG

Pinecone vector store (ErayaGraph) · CyborgDB encrypted memory · Firecrawl ingestion.

Quant trading engine

Live Gate.io / KuCoin / CoinGecko candles · SMA/RSI/Bollinger/momentum + Fear & Greed (contrarian) ensemble · vol-targeted sizing · voice risk dial (Whisper in, ElevenLabs out).

Economic signal

x402 prices inter-agent work — external agents buy the swarm's live quant signal over HTTP 402 (swarm-for-hire).

Threats hunted before a bad deploy lands.

KAVACHA

Security-first guardian for unsafe actions, adversarial inputs, and agent identity.

Live attack sim

verdict BLOCKED · rule R003 · full signed timeline

Signed audit trail

HMAC-signed timeline — every veto recorded

DeFi policy rules R004–R008

mempool · liquidity · governance · slippage

Injection kill-shot

Operator/agent inputs scanned; unsafe actions BLOCKED with a signed audit trail.

A2A identity defense

Forged agent messages rejected via HMAC signature check — same function the live WebSocket path uses.

Quorum before execution

High-stakes DeFi moves need swarm consensus, not one agent’s call.

Proactive threat scanner

Mempool, liquidity, and governance risk scanned before execution — DeFi policy rules R004–R008.

The live deployment serves the console right now — with real on-chain execution.

<https://eraya.34.126.112.227.nip.io> — HTTPS · sign in as Guest

4,997

CSPR TREASURY · LIVE RPC

2

ODRA CONTRACTS DEPLOYED

4

AGENTS REGISTERED ON-CHAIN

2.5

CSPR SETTLED PER TRADE

TabPFN

RISK SCORE

200

DASHBOARD API

Live system surfaces respond
KAVACHA attack-sim, swarm status, live A2A reasoning chat, and the voice copilot all respond with real data.

Production, not slideware
Real testnet transactions: contract deploys, agent registrations, trade settlement — every one with a cspr.live link.

Swarm-for-hire over x402
External agents — elizaOS, LangChain, MCP, curl — pay 0.005 CSPR over HTTP 402 for the swarm's live quant signal.

One engine, three of Casper’s priorities.

DeFi

Autonomous treasury rebalance, yield monitoring, and slippage/liquidity-aware execution.

RWA-ready

The swarm is domain-agnostic — a 3-method adapter plugs in any real-world asset stream, already proven across 5G, cloud, and ICU domains.

Agentic AI

Self-healing, quorum, priced agent economy — the “how do agents act safely with real money” story.

DeFi × RWA × AI

Defined failure behaviour is our moat.

Every agent has a defined failure path

The swarm never fully stops — it can re-plan, retry, roll back, or fail safely.

Live, open-source, production-ready

Judges can hit the URL now and inspect the code.

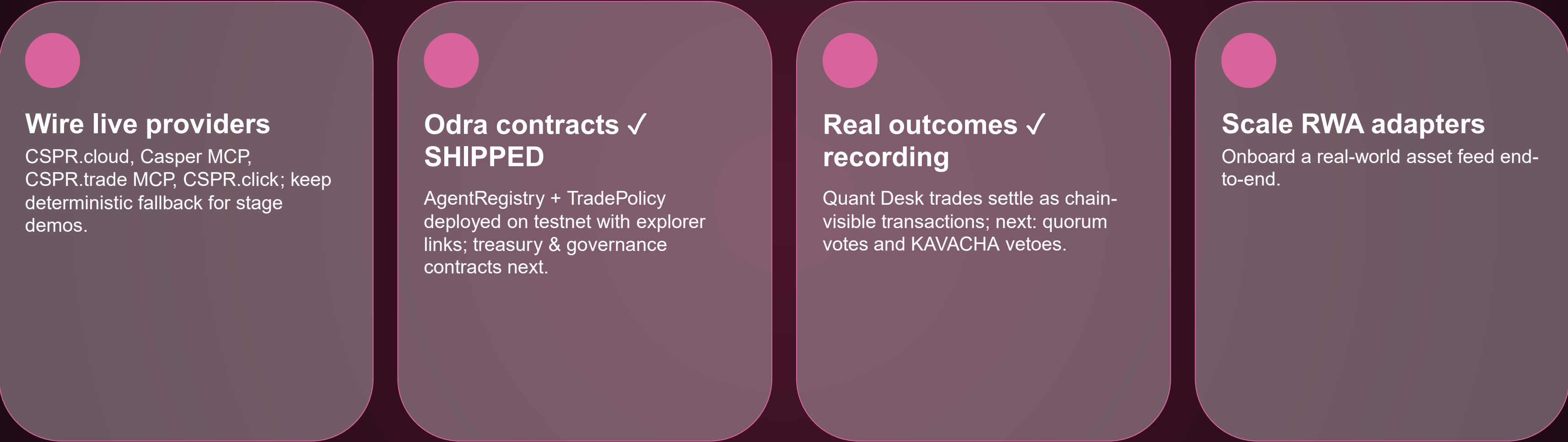
Real signals on the critical path

Live market data, deployed Odra contracts, real testnet settlement — no critical-path stubs.

Two hard problems solved together

Agent safety through KAVACHA + agent economics through x402 + reputation.

Contracts and settlement are LIVE; next: providers and RWA.



Execution path: Odra contracts ✓ → on-chain outcomes ✓ → live providers → RWA adapters

Team ERAYA

Team Lead

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Built end-to-end: agents, Odra contracts, quant desk, voice.

Live demo

<https://eraya.online>

Guest / member / admin sign-in — trading, wallet, KAVACHA, A2A feed, x402, voice.

Role allowances: Guest 30 CSPR/tx · member 100 · admin unlimited.

Repository

github.com/martian3062/eraya
branch caspr · open-source

Odra 2.9 · 7 MockVM tests · one-command deploy script.

The ask

Final Round: judge the live system — a self-healing agentic DeFi treasury with deployed Odra contracts and real on-chain trading, built Casper-native.